Philip Bulsink

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Summary

Senior analytical chemist with over a decade of experience leading advanced chromatographic and spectroscopic method development to drive biofuel and pyrolysis oil characterization. Proven track record of leading international interlaboratory studies, spearheading novel analytical approaches for complex biorefinery matrices, and authoring peer-reviewed publications. Experienced in collaborating across multi-disciplinary teams of RES scientists, research engineers, and technologists to transform complex analytical data into actionable research insights.

Current Position

Fuels Chemist (CH-03) — October 2014 — Present

Characterization Laboratory, CanmetENERGY-Ottawa, Natural Resources Canada — *Ottawa, Ontario*

- Perform detailed chemical and physical analyses of petroleum- and bio-derived fuels and liquid research products using advanced chromatographic and spectroscopic techniques.
- Pioneer analytical methods using custom-built chromatographic and spectroscopic equipment (GC-FID/MS, GCxGC-FID, GCxGC-FID/MS, FTIR) to characterize biofuels and petroleum fuels, resolving complex analytical bottlenecks for research groups and industry partners.
- Engineer custom chemometric algorithms and open-source R tools to automate complex spectroscopic data analysis and visualization across research projects.
- Process and interpret high-dimensional analytical data to deliver actionable insights and support fuel production research scientists.
- Lead and co-author journal publications and conference presentations, contributing to the dissemination of research findings.
- Contribute to research funding proposals, strengthening experimental design and improving project outcomes.
- Represent the laboratory, department, and Canadian scientific interests at internal, national and international fora, including conferences, technical meetings, working groups, and standards boards.
- Spearheaded international round-robin studies of bio-liquefaction oils, fostering collaboration and advancing scientific understanding of analytical methods.
- Champion data quality systems by serving as a CALA-certified ISO 17025 internal auditor, verifying complex analytical processes to ensure strict compliance and experimental reliability.
- Train and support junior staff and co-op students, fostering a culture of knowledge sharing and professional development within the laboratory.
- Act up as CH-04 with managerial and Sections 32 & 34 authority, as required.

Key Projects and Activities

Method Development Lead, SPME Analysis of Biorefinery Products — 2025 — Present

- Developing analytical methodology (using Design of Experiment techniques) for analysis of products produced in a biorefinery, regardless of matrix.
- Achieves high sensitivity and good robustness for key carbonyl target compounds impacting employee health and safety.

Developer & Maintainer, PlotFTIR (https://nrcan.github.io/PlotFTIR) — 2024 — Present

- Developed an R package for plotting and analyzing Fourier Transform Infrared (FTIR) spectroscopy data, enabling researchers to visualize and interpret complex spectral information effectively.

Method Development Lead, GC-FID/MS Semiquantitative Analysis — 2018 — 2022

- Built chromatographic method and custom data analysis algorithms to produce semiquantitative compositional reports with nearly 100% mass balance for complex samples. This highly robust technique supports petroleum, low carbon/renewable fungible fuels, and bio-fuel product and process development to the present day.

Emergency Response Team — 2015 — Present

- Comprehensively trained in emergency medical and chemical response procedures, including advanced first aid, burn response, trauma/wound care, emergency triage, scene control and hazardous spill control operations.
- Act as backup dispatch, incident commander, and technical HAZMAT expert.
- Maintain annual SCBA recertification.

Previous Experience

Research Assistant (Co-Op) 2010 — 2011
DeNOx Group, CanmetENERGY-Ottawa, Natural Resources Canada *Ottawa, Ontario*

- Investigated homogeneous catalysts for the reduction of NO_x in lean-burn diesel engine exhaust.
- Scaled catalyst synthesis by three orders of magnitude.
- Custom-built analytical instrumentation and analysis software to support research inquiries.

Laboratory Teaching Assistant September 2011 — May 2014
University of Ottawa / University of Waterloo *Ottawa / Waterloo, Ontario*

- Supervised undergraduate laboratories across analytical, organic, inorganic, and physical chemistry, instructing on advanced techniques and lab safety protocols.

Education

University of Ottawa 2012 — 2014
Master's of Science, Chemistry *Ottawa, Ontario*

- Thesis: "Rhenium (I) Terdentate Compounds: Theoretical and Experimental Investigations"
- Seminar: "Recent Advancements in NO_x Abatement from Diesel Engine Emissions"

University of Waterloo 2007 — 2012
Bachelor's of Science, Honours Chemistry (Co-op), Music Minor *Waterloo, Ontario*

- Thesis: "Solid Sample Analysis by Microplasma Optical Emission Spectroscopy"
- Achieved "Outstanding" co-op term ranking for each of the 5 work terms (2008-2012)

Extracurriculars & Community Leadership

Clerk of Council (Executive Committee), Kanata Community Church 2024 — Present

- Provide executive governance and administrative leadership for the church, managing council records, overseeing committee logistics, and supporting strategic operations and policy execution.

Author & Maintainer, f1dataR R Package 2023 — Present

- Developed and maintain an open-source R package for F1 data analytics, managing CRAN submission requirements, automated test coverage workflows, and community issue tracking on GitHub.

Awards and Distinctions

Scientific Innovation, Branch Award, CanmetENERGY-Ottawa 2026
Excellence in Science, Departmental Award, Natural Resources Canada 2023
Positive Workplace Impact, Sector Award, Energy Efficiency & Technology 2022
Innovation & Creativity, Branch Award, CanmetENERGY-Ottawa 2021
Dean's Scholarship, University of Ottawa 2015
Dean's Honour Roll, University of Waterloo; University of Ottawa 2011-2014
Recognition of Collaboration, Departmental Award, Natural Resources Canada 2012
Aileen Proudfoot Award, Branch Award, CanmetENERGY-Ottawa 2011

Technical Skills & Expertise

- **Coding & Data Analysis Tools**:
  - R (package development, CRAN maintenance), Python, VBA, RDKit
  - Agilent MassHunter, Agilent Chemstation, Bruker TopSpin, Gaussian, TurboMol
- **Analytical Instrument Experience**:
  - Agilent Gas Chromatography Systems, including PAL autosamplers with liquid, headspace, and SPME sampling; FID, mass spectral, FPD, PFPD, and TCD detectors; and custom multi-column setups with CFT Flow splitters or GCxGC flow valves
  - FTIR instruments (ATR 5-bounce and 1-bounce systems, DRIFTS)
  - NMR spectroscopy, HPLC, UV-Vis, elemental analyzers, titrators, and other standard laboratory instrumentation
- **Linguistic Profile**: English native | French CBA

Selected Publications and Presentations

- **Bulsink, P.**; *et al.* (2026) “Characterization of Volatile Carbonyl Compounds in Biorefinery Streams using SPME-GC: Implications for Occupational Health and Safety” **Pyro2026 Conference**. Pisa, Italy
- Zhang, Y.; Monnier, J.; **Bulsink, P.**; *et al.* (2025). “Evaluation of diesel fuel production from bio-oils hydrodeoxygenation using unsupported MoS_2 catalysts” **Fuel Processing Technology**, 276, 108290. 10.1016/j.fuproc.2025.108290.
- **Bulsink, P.**; *et al.* (2025). “Multidimensional Chromatography of Bio/Renewable Energy Products “ **Canadian Society of Chemistry Conference**. Ottawa, Ontario
- **Bulsink, P.**; *et al.* (2025). “Interlaboratory Study of Sample Homogeneity Impact on CHNS, Water, and ICP Analysis of Biomass Liquefaction Oils” **Energy & Fuels**, 39 (29), 14223-14236. 10.1021/acs.energyfuels.5c01309.
- **Bulsink, P.**; *et al.* (2023). “Quantification of components without direct calibration by GC-MS/PolyArc®-FID” **American Chemical Society Conference**. San Francisco, California
- **Bulsink, P.** (2021). “Gas Chromatographic and Mass Spectral Analysis in Characterization Lab” *CanmetENERGY-Ottawa Science Seminar*
- **Bulsink, P.**; *et al.* (2020). “Results of the IEA Bioenergy Round Robin on the Analysis of Heteroatoms in Biomass Liquefaction Oils” *CanmetENERGY-Ottawa Science Seminar*
- **Bulsink, P.**; *et al.* (2020). “Results of the International Energy Agency Bioenergy Round Robin on the Analysis of Heteroatoms in Biomass Liquefaction Oils” **Energy & Fuels**, 34 (9), 11123-11133. 10.1021/acs.energyfuels.0c02090.
- **Bulsink, P.**; *et al.* (2016). “Capturing Re(I) in an neutral N,N,N pincer Scaffold and resulting enhanced absorption of visible light” **Dalton Transactions** 45, 8885-8896. 10.1039/c6dt00661b.
- Stanciulescu, M.; **Bulsink, P.**; *et al.* (2014). “NH_3-TPD-MS study of Ce effect on the surface of Mn-or Fe-exchanged zeolites for selective catalytic reduction of NO_x by ammonia” **Applied Surface Science** 300, 201-207. 10.1016/j.apsusc.2014.01.175.
- Stanciulescu, M; Caravaggio, G.; Dobri, A.; Moir, J.; Burich, R.; Charland, J.-P.; **Bulsink, P.**; (2012). “Low-temperature selective catalytic reduction of NO_x with NH_3 over Mn-containing catalysts” **Applied Catalysis B: Environmental** 123, 229-240. 10.1016/j.apcatb.2012.04.012.

Professional Affiliations

- ASTM International — 2024 — Present
- American Chemical Society — 2019 — 2026
- CGSB Petroleum Fuels Board — 2016 — 2025
- Canadian Society for Chemistry — Periodic Member, 2013 — 2025